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Comparative genomics of ocular *Pseudomonas aeruginosa* strains from keratitis patients with different clinical outcomes

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ABSTRACT

The several virulence and drug-resistance mechanisms of *Pseudomonas aeruginosa* responsible for poor clinical outcomes in keratitis patients remain largely unknown. Here, we investigated the distribution of virulence factors and drug resistance by genes, mutations, efflux-pump systems of *P. aeruginosa* strains from keratitis patients with different clinical outcomes, included of whole-genome sequenced and annotated our five *P. aeruginosa* strains. Of the large number of virulence genes detected in all the genomes, MDR/XDR strains carry exoU and non-MDR strains carry exoS exotoxin of the type III secretion system, considered as main contributors of keratitis pathogenesis. However, several strain-specific virulence and resistance genes were detected in keratitis strains with poor outcome. Mainly, the flagellar genes fliC and fliD detected in the exoS carrying strains, reported to alter the host immune response, might impact the clinical outcome. This study may provide new insights into the genome of ocular strains and requires further functional studies.

1. Introduction

Pseudomonas aeruginosa is a destructive opportunistic pathogen responsible for bacterial keratitis characterized by ocular pain, infiltration of inflammatory cells, stromal destruction, corneal perforation and leads to vision loss if untreated [1]. Keratitis pathogenesis is a complex process, wherein several virulence factors have been implicated including cell-associated structures such as type IV pili and flagella [2,3], slime polysaccharide, proteases such as elastase B (LasB), alkaline protease (AprA), protease IV (PrpL) and *P. aeruginosa* small protease (Pasp), and exotoxins [4]. However, several studies revealed that exotoxins exoU and exoS that belong to the type III secretion system (T3SS) are important contributors to keratitis pathogenesis [5,6]. *P. aeruginosa* strains expressing the exoS gene are known to invade epithelial cells, whereas strains expressing the exoU gene, a potent phospholipase, are cytotoxic towards host cells [7,8]. ExoU strains have been reported to cause more severe infection [9] than exoS. Moreover, clinical strains from keratitis prefer to carry exoU gene rather than exoS in their accessory genome [5,10], although the core genome (approximately 90% of the complete genome) contains most of the virulence genes [11]. Besides, the clinical strains carrying multiple antimicrobial drug resistance in the accessory genome are associated with poor clinical outcome in patients with keratitis [12]. Indeed, multiple drug-resistant (MDR) strains carrying exoU protein may lead to poor clinical outcomes [12]. Furthermore, a subpopulation of UK-keratitis associated *P. aeruginosa* strains have been reported to have specific features that cause eye infections [10]. However, the link between specific genomic features and clinical outcome remain unclear.

In this study, we have selected five strains cultured from the keratitis patients based on their drug resistance profiles, the presence of T3SS virulence factors, and clinical outcome. Here, we report the analysis of whole-genome sequenced and annotated genomes of five *P. aeruginosa* ocular strains compared to reference genome PAO1 and compared them with sixteen other genomes of *P. aeruginosa* keratitis strains. We explore how the changes in molecular features that impact the clinical outcome of the keratitis patients by investigating the association between virulence, drug-resistance and clinical outcome of the patients.

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Table 1
Minimum inhibitory concentration and clinical data of *P. aeruginosa* ocular strains of the current study. R, resistant; I, intermediate; S, susceptible.

Antibiotic classes	Antibiotics	BK2	BK3	BK4	BK5	BK6
Penicillin and beta-lactamase inhibitor	Ticarcillin/clavulanic acid	R ≥ 128	R ≥ 128	*R* ≥ 128	R ≥ 128	R ≥ 128
	Piperacillin/Tazobactam	R ≥ 128	R ≥ 128	R ≥ 128	R ≥ 128	R ≥ 128
Cephalosporins	Ceftazidime	S 2	S 4	S 2	S 2	R 32
	Cefepime	R ≥ 64	R ≥ 64	R ≥ 64	*R* ≥ 64	R ≥ 64
Carbapenems	Doripenem	S 1	*R* ≥ 8	I 4	S 1	R ≥ 8
	Imipenem	S 2	*R* ≥ 16	S 2	S 2	*R* ≥ 16
	Meropenem	S	I 4	S	S	R 4
Aminoglycosides	Amikacin	S 4	R 16	S 4	S 4	*R* > 64
	Gentamicin	S 2	*R* > 16	S 2	S 2	R > 16
	Tobramycin	S 1	R ≥ 64	S 1	S 1	R ≥ 64
Fluoroquinolones	Ciprofloxacin	S < 0.25	S 0.5	S 0.5	S < 0.25	*R* > 4
	Levofloxacin	S 1	S 1	S 1	S 1	*R* > 8
	Moxifloxacin	S 1	S 2	S 2	S 1	R ≥ 64
	Gatifloxacin	S 0.5	S 2	S 1	S 0.5	R ≥ 64
Polymyxins	Colistin	S 2	S 2	S 2	S 2	S 1
Clinical outcome[a]		Healed	TPK	TPK	TPK	Healed
Patient age/sex		31/M	65/M	60/M	68/M	50/F
T3SS-type		ST	UT	ST	UST	UT

Minimum inhibitory concentration (MIC) was determined using VITEK2 and broth dilution method. The values are represented in μg/mL.
[a] TPK- Therapeutic Penetrating Keratoplasty. T3SS- Type three secretion system genotyping.

2. Materials and methods

2.1. Ocular bacterial strains

Five *P. aeruginosa* strains obtained from keratitis patients attending Aravind Eye Hospital, Madurai, India, were selected retrospectively. The Institutional Review Board approved the study protocol and informed consent was obtained from each patient before sampling. The study was performed according to the tenets of the Declaration of Helsinki. Five strains were selected based on clinical outcomes and antibiotic sensitivity pattern (Table 1). Two strains (BK2 and BK6) cultured from the corneal scrapings of patients who subsequently presented with a healed ulcer after treatment were categorized as good clinical outcome group. The other three strains (BK3, BK4 and BK5) cultured from corneal buttons from patients who underwent therapeutic penetrative keratoplastic (TPK) surgery were categorized as poor outcome group. Bacterial identification was performed based on morphology and growth characteristics in culture, gram's staining and 16 s rDNA sequencing. The strains were stored at −80 °C at the microbiological culture collection facility.

2.2. Antimicrobial susceptibility testing

The Minimum Inhibitory Concentration (MIC) of various antibiotics were determined using the broth microdilution method and automated BioMérieux VITEK 2 system following the recommendations of the Clinical and Laboratory Standards Institute (CLSI, 2012). *P. aeruginosa* ATCC® 27853™ strain was used for quality control. Briefly, in the broth dilution method, the antibiotic solutions were serially diluted and inoculated with test organisms in Mueller-Hinton broth and incubated for 18–24 h at 37 °C. The lowest concentration of an antibiotic that prevented bacterial growth (turbidity) was considered as the MIC. For VITEK based determination, the colonies grown on blood agar for 18–24 h at 37 °C were resuspended and adjusted to 0.5 McFarland standard with 0.45% saline. The reagent cards (AST−N281) were inoculated with culture suspensions according to manufacturer's instructions and data was collected during the entire incubation period.

Table 2
Genomic features and statistics of ocular *P. aeruginosa* strains of this study.

Strain	BK2	BK3	BK4	BK5	BK6
Genome size (Mb)	6.3	7.1	6.5	6.3	7.1
G + C	66%	65%	64%	66%	66%
Scaffolds	45	113	69	68	129
Maximum length (bp)	703,333	594,831	700,326	877,303	602,407
Minimum length (bp)	506	504	509	509	510
N50	370,564	191,913	347,365	350,169	232,253
CDs	6037	7009	6141	6055	6722
Hypothetical proteins	1042	1460	1079	1078	1327
tRNA	61	65	60	61	60

2.3. Type 3 secretion system (T3SS) genotyping

T3SS genotypes were determined by PCR using specific primers *exoS*, FP (5′CCCGGAACTGCGGCGCGAAATCAC3′); RP (5′CCGGT TTG CTTGCCAGGTCGAGAG3′); *exoT*, FP (5′ACTCCCCGGGAGGCGCAAC AAC3′) RP (5′CTTGTCGACAGGCTCGCCCTTT3′) and *exoU*, FP (5′CAG GAGCGAGTCGGTG AACATCTG3′); RP (5′CGCCCGACGTAACAGAGC TAC3′) against exoS, exoT and exoU genes. Briefly, 25 μl PCR reactions were set up with 1× reaction buffer, 200 nM primers, 400 μM dNTP and one-unit Taq polymerase. The reaction conditions consisted of an initial denaturation step at 94 °C for 10 min, followed by 35 cycles of 94 °C for 2 min, 60 °C for 30 s and 72 °C for 1 min, with a final extension at 72 °C for 5 min. The reference strains PAO1 and PA14 served as positive controls.

2.4. DNA isolation and genome sequencing

Genomic DNA was isolated from strains using the QIAamp DNA mini kit (Qiagen, Hilden, Germany) according to their manufactures protocol. NanoDrop spectrophotometer (Thermo Scientific, Wilmington, USA) was used for purity and concentration of DNA. Whole-genome sequencing (WGS) was performed at Genotypic, India. Briefly, the genomic DNA was fragmented to obtain 200 to 600 bp size by Covaris S220. Paired-end libraries (∼140–480 bp insert size) were prepared using NEXTFlex DNA sequencing kit and quantified with Bioanalyzer (Agilent). WGS was performed using Illumina NextSeq 500 platform and the raw data of each strain were obtained through demultiplexing.

2.5. Genome assembly and analysis

The quality of the paired-end sequence was measured by FastQC (version 0.11.4). The adapter sequences, low coverage reads and PCR duplicates were excluded before the assembly, using cutadapt3 version 1.9.1 [13]. The high-quality reads were further assembled into contigs with CLC genomic workbench (version 9.0.1) with the default setting. Contigs ordering and visualizations were performed using Mauve program version 2.3.1 [14] using the PAO1 genome as a reference (RefSeq accession number NC_002516.2). The assembly was further improved using Genome Finishing Module in the CLC genomics workbench. Rapid Annotations Subsystems Technology (RASTtk) [15] server was used for annotation (Table 2).

2.6. Comparative genomics

We compared the genomes of this study with 16 reported *P. aeruginosa* genomes from keratitis patients [10,16–18] and reference strains PAO1 [19] and UCBPP-PA14 [20] [Supplementary Table S1]. The annotated genomes were downloaded from GenBank database. Parsnp version 1.2 [21] was used to align the genomes of *P. aeruginosa* strains, followed by maximum-likelihood phylogenetic tree construction based on core-genome SNPs using PAO1 as a reference genome. The maximum-likelihood phylogenetic tree was visualized by iTOL version 4 [22]. The default parameters were used to generate the tree with *P. aeruginosa* PA7 [23], a taxonomic outliner, as an outgroup.

Roary version 3.12.0 [24] was used to perform pangenome analysis to identify the number of core, dispensable and strain-specific genes using the GFF3 files with the default setting. The extent of accessory genomes of all the strains were plotted using Circos [25] and phandango [26]. MLST prediction server [27] was used to identify sequence types of each strain. ResFinder version 3.2 [28] was used for the detection of antibiotic resistance genes and 90% threshold and 80% of minimum length were set. BLAST Ring Image Generator (BRIG) [29] tool with 90% upper and 70% lower sequence identity and *E*-value cutoff of 1e-5 was used for virulence gene identification against Virulence Factor of Pathogenic Bacteria database (VFDB) [30]. PHASTR server was used for complete prophage identification [31].

2.7. Variant calling

The paired-end raw reads of all five isolates were mapped with PAO1 reference genome using BWA version 0.7.17 [32] with default parameters. The reference mapped reads were pre-processed, such as the removal of PCR duplicates using Picard version 2.23.3 [http://broadinstitute.github.io/picard/]. Then, GATK version 4.1.4.1 [33] was employed to call the variants.

2.8. Validation by sanger sequencing

Bi-directional sequencing method was performed to validate the mutations in genes that are associated with antibiotic resistance. The primers sequences, annealing temperatures, and amplicon sizes are mentioned in Supplementary Table S2. A thermal gradient PCR was performed with 120 nM primers, 400 µM dNTP and one-unit Taq polymerase in a 25 µl reaction. 3130 Genetic Analyzer (Applied Biosystems) was used for sequencing.

2.9. Motility assay

2.9.1. Swimming

Swim plates (1% tryptone, 0.5% NaCl, 0.3% agar) were inoculated with 0.2 µl of overnight cultures and incubated for 24 h at 30 °C. The turbid zone formed by the bacterial cells migrating away from the point of inoculation was assessed.

2.9.2. Swarming

0.2 µl of overnight cultures were spotted on the surface of the swarm plates (0.5% Luria Bertani (LB) agar and 0.5% glucose) and incubated for 24 h at 30 °C.

2.9.3. Twitching

1 µl of overnight cultures were stabbed into the 1.5% LB agar plates until the culture reaches the surface of the plate. After 24 h of incubation at 37 °C, a hazy zone of growth at the interface between the agar and the polystyrene surface of the plate was assessed by the crystal violet staining (1% [wt/vol] solution) after the removal of the agar. PAO1 and PA14 were used as controls.

2.10. Data access

The whole-genome sequencing data of all five strains of this study is available at NCBI-SRA with accession number SRR2062214, SRR2063971, SRR2063976, SRR2063980 and SRR2063985.

3. Results and discussion

3.1. Antibiotic susceptibility profile and genome characteristics

Antibiotic susceptibility profiling (Table 1) was carried out based on the common clinical use of anti-pseudomonal drugs for keratitis [34,35]. Multi-drug resistance (MDR) is defined as non-susceptibility to at least one agent in ≥3 classes of antibiotics and extensively drug-resistant (XDR) is defined as non-susceptibility to at least one agent in all but two or fewer antimicrobial categories [36]. BK3 and BK6 strains were identified as MDR and XDR strains, whereas BK2, BK4 and BK5 were identified as non-MDR phenotype as shown in Table 1. Of all the keratitis strains included in this study (Supplementary Table S1), six non-MDR, five MDR, and four XDR strains were identified except PA39016 due to lack of clear antibiotic tested results. All the strains were sensitive to Colistin. MDR/XDR (high resistance profile) strains were found to have a bigger genome size of ~6.8 MB and more genes compared to ~6.3 MB of non-MDR strains. Nevertheless, the poor outcome (BK1, BK3, BK4, BK5 and VRFPA04) or good outcome group (BK2 and BK6) strains were found to have different resistance profiles.

3.2. Pattern of gene distribution

Nine thousand and two orthologs were detected in five genomes of this study and PAO1 compared to 9344 orthologs of reported *P. aeruginosa* genomes [37]. Among the detected orthologs, we identified 5139 genes as the core genome. The pattern of accessory, strain-specific and overlapping genes among the five strains are shown in Supplementary Fig. S1. Furthermore, pangenome analysis of all the keratitis strains with PAO1 detected 12,775 orthologs in total (Supplementary Fig. S2). Among them, 4196 core genes and ~ 1920 accessory genes were detected in each strain. Among the poor outcome group, BK3 strain shared more accessory genes with PA31, PA32, PA33, PA35 and PA37, while BK1, BK3 and VRFPA04 had more unique (strain-specific) genes, which were mostly hypothetical proteins.

The analysis of Clusters of Orthologous Groups (COG) of core and unique genes showed that the core genes were enriched to class E (Amino acid transport and metabolism) and class T (Signal transduction mechanisms) (Supplementary Fig. S3). The unique genes were enriched to class L (Replication, recombination and repair), class M (Cell wall/membrane/envelope biogenesis), class I (Lipid transport and metabolism) and class S (Function unknown), suggesting that they may help for bacterial survival.

3.3. Phylogenetics

We compared phylogenetic diversity of the five strains of this study with other *P. aeruginosa* keratitis strains including two reference strains PAO1 and UCBPP-PA14. Based on the core genome SNP phylogenetic tree, the strains were clustered into two groups. One group consisted of only non-MDR strains and another group consisted of MDR/XDR (Fig. 1). Among MDR/XDR group, BK3 of this study was closely related

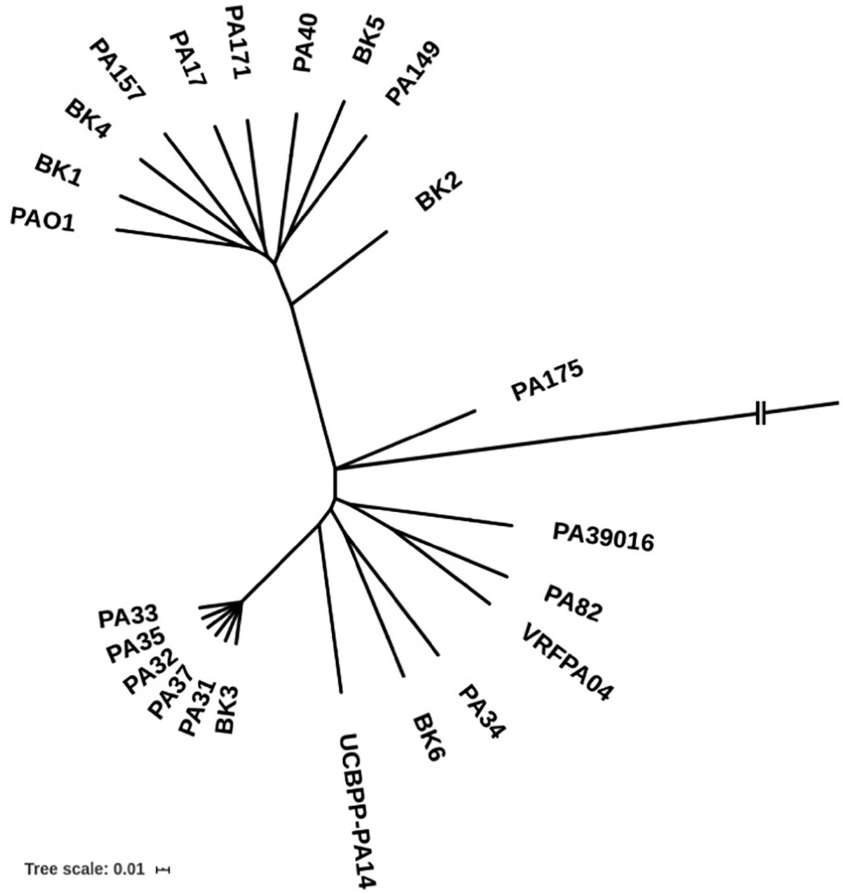

Fig. 1. Phylogenetic tree of *P. aeruginosa* strains. Maximum likelihood tree was created based on core genome SNPs mapped to the reference genome PAO1.

to strains (PA31, PA32, PA33, PA35 and PA37) isolated from India and BK6 was closely related to UCBPP-PA14. The other strains in this group were PA39016 isolated from England, PA175 from Australia, VRFPA04 and PA82 from India. While the non-MDR strains (BK2, BK4, BK5) were closely related to all other non-MDR strains (PA17, PA40, PA149, PA157 and PA171) from Australia and BK1 from India. However, either the strains from good or poor outcome patients were not clustered into the same group based on the phylogenetics.

3.4. Antibiotic resistance

3.4.1. Resistance genes

We detected twenty antimicrobial associated resistance genes in all five genomes of this study. Seven genes were present in all five, as shown in Fig. 2. Among them, the β-lactamase genes blaPAO and blaOXA-50, are known to cause resistance to the carbapenem class of antibiotics [38,39] and aph(3′)-IIb gene may provide resistance to many aminoglycoside class of antibiotics. blaPAO and aph(3′)-IIb was commonly present in all keratitis strains and blaOXA-50 were mostly present in all keratitis strains except PA39016 and VRFPA04 (Fig. 2). The catB7 gene of CATB family reported to cause chloramphenicol resistance [36] and fosA gene to Fosfomycin, were also present in all the keratitis strains except BK1 and VFRPA04, respectively. Interestingly, pmrA-pmrB gene that provides resistance to polymyxin B was present only in the strains (BK2-BK6) of this study. As expected, we observed more resistance genes (17) in MDR/XDR than non-MDR strains (11) of this study. However, the other keratitis strains carried different resistance gene profile for their resistance to multiple drugs as shown in the Fig. 2. The tetG gene, responsible for tetracycline resistance, was detected in all the XDR/MDR strains expect PA34 and PA82. Besides, we observed several strain-specific genes in this study; rmtA and rmtB (16S rRNA methylase genes), rosB and drfB5 in BK6, strA and strB (streptomycin-inactivating enzymes) in BK3, and PDC-5 in BK5. However, their role in the clinical outcome of the patients is not known.

3.4.2. Gene mutations associated with drug resistance

To identify drug resistance-associated point mutations leading to amino acid change, we detected mutations in drug target genes or mutations that overexpress resistance protein using PAO1 as reference (Fig. 3). We detected several mutations in ampR and ampC genes. Among them, G27D, R79Q, T105A, Q155R, V205L, V356I, and G391A were reported to overexpress intrinsic ampC protein [40,41], and thus they provide penicillin and cephalosporin resistance to ocular strains. Of these, T105A was common in all keratitis strain. Two mutations V205L and G391A were present only in exoU strains. Furthermore, two mutations G283E and M288R in ampR were mostly present in all keratitis strains, except exoS strains but BK4 and BK5. Also, several mutations were identified in fluoroquinolones target genes gyrA (DNA gyrase) and parC (Topoisomerase IV). Of these mutations, T83I of gyrA mutation (T83I) and parC mutation (S87L) detected in BK6 were previously reported to cause resistance to fluoroquinolones [42–44]. In addition, one strain-specific gyrA mutation was detected in BK2 in this study. Studies report that mutations in gyrA along with mutations in parC might provide the higher-level resistance to fluoroquinolones [42,45]. Indeed, the combination of gyrA (T83I) and parC (S87L) mutation was observed in MDR/XDR keratitis strains carrying ExoU except PA34, PA82 and PA175, which were associated with the higher MICs for fluoroquinolones (Supplementary Table S1) [46]. Furthermore, we detected oprD gene mutation in all strains of this study (except BK3) and other keratitis strains as shown in Fig. 3, which may confer resistance to carbapenems. However, these mutations are not associated with carbapenems resistance profile of the strains (Supplementary

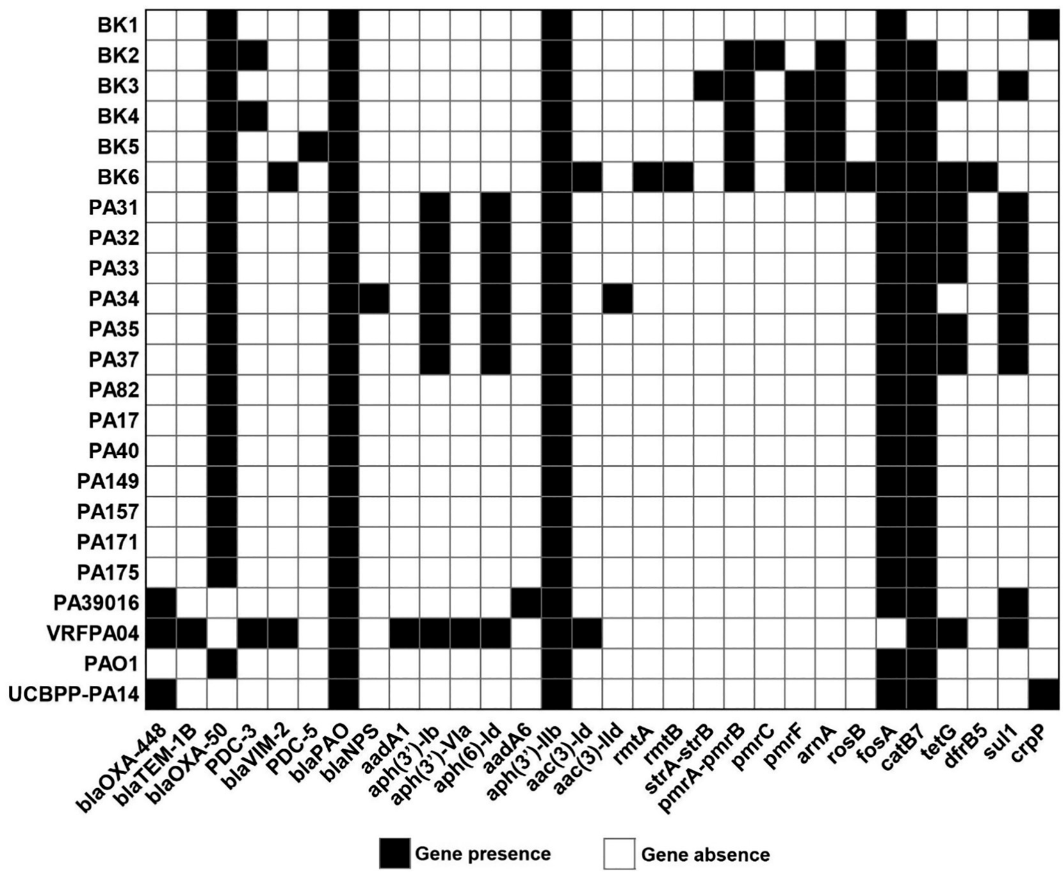

Fig. 2. The presence and absence of antimicrobial resistance genes detected in *P. aeruginosa* keratitis strains and reference strains PAO1, UCBPP-PA14.

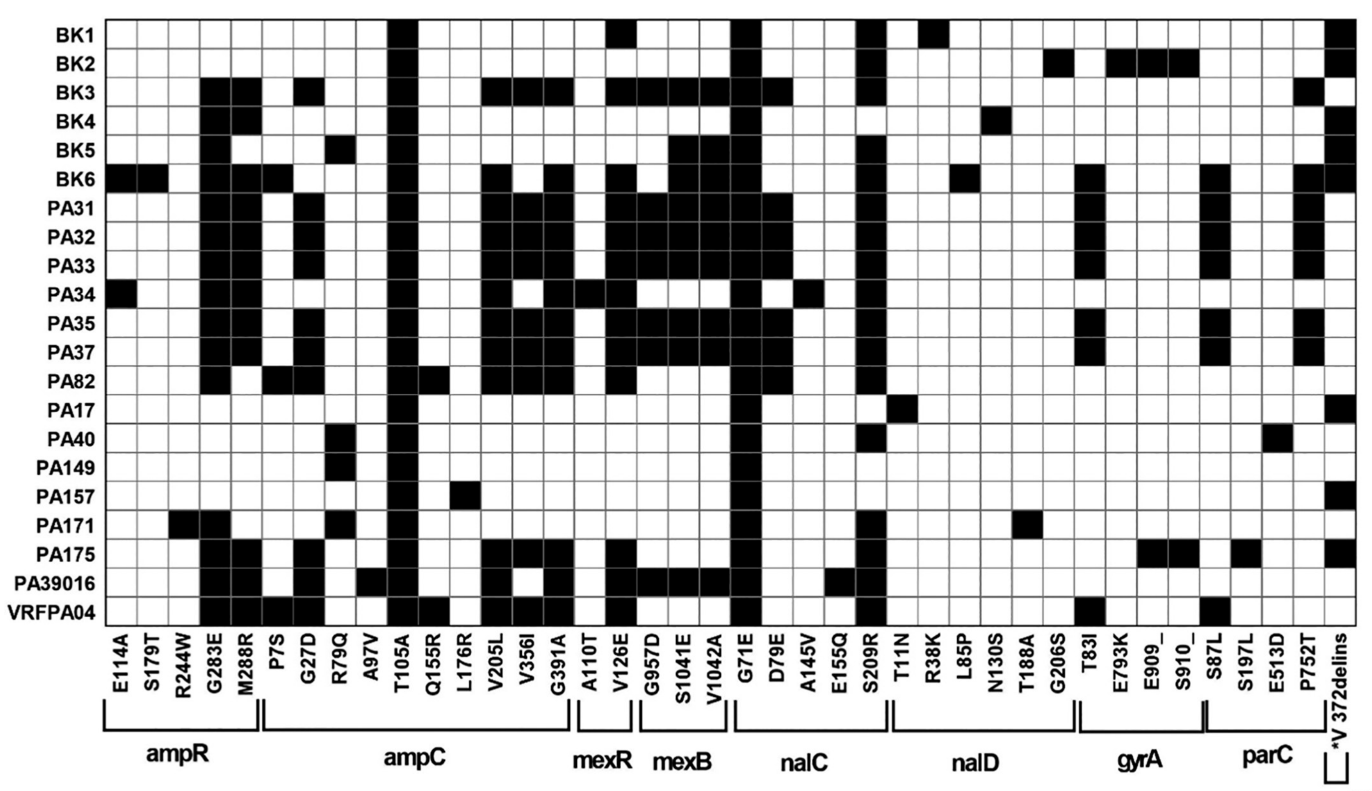

Fig. 3. Antibiotic resistance-associated mutations in genes in *P. aeruginosa* keratitis strains compared to PAO1 as reference genome. *V 372delins = 372 VDSSSS--YAGL 383. Black box represents the presence of mutation and white represents absence.

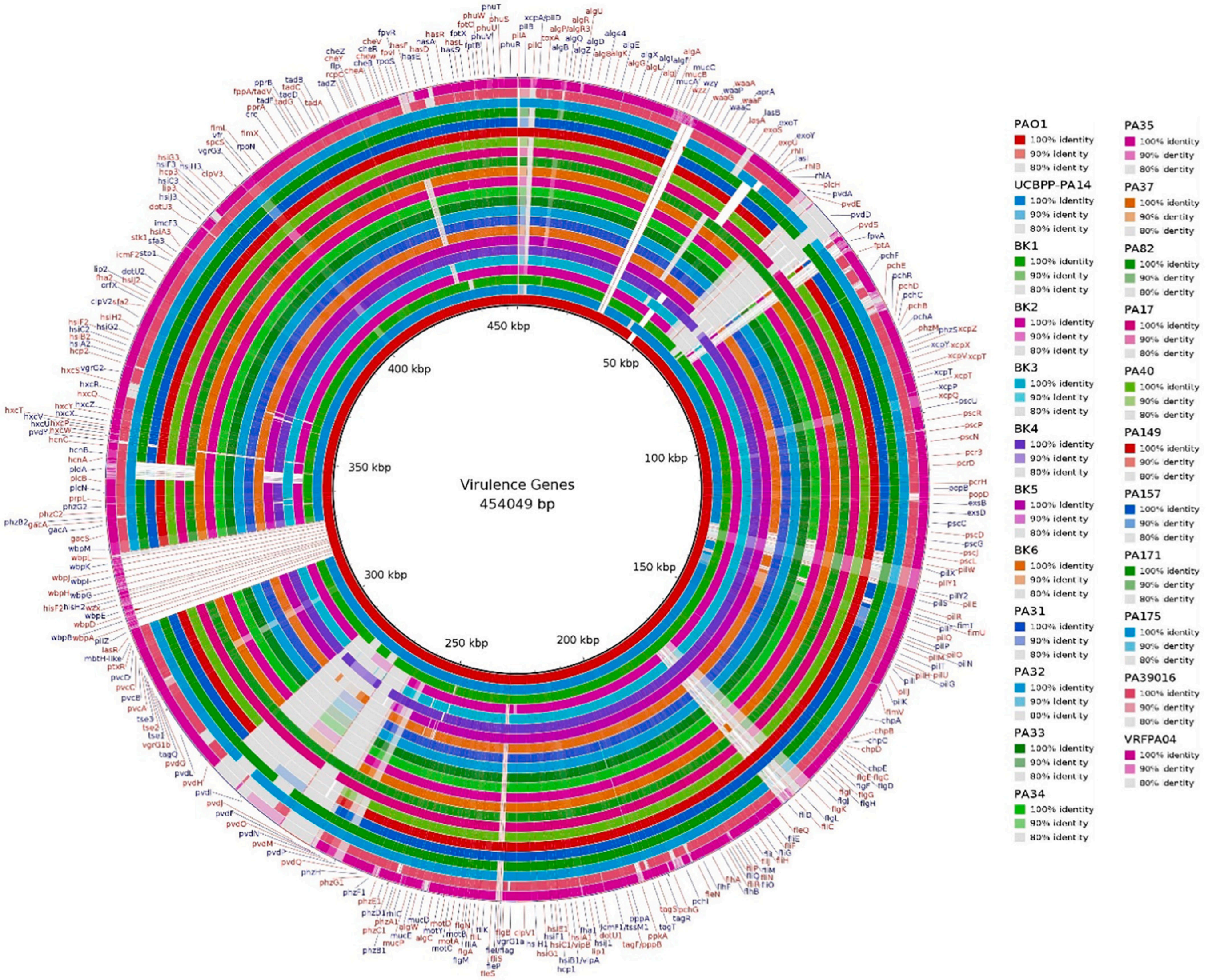

Fig. 4. A BLAST ring representation of virulence genes detected in *P. aeruginosa* keratitis strains and reference strains PAO1, UCBPP-PA14. The color coding indicates the sequence identity in percentage against VFDB database. Image was created using BRIG tool.

Table S1).

Genes of multiple efflux pump systems, MexAB-OprM, MexCD-OprJ and MexEF-OprN, which belongs to RND (resistance-nodulation-division) efflux pump family, were identified in the strains of this study. However, we detected several non-synonymous point mutations in transcriptional regulator genes mexR, nalC nalD, periplasmic membrane fusion protein mexA and inner membrane drug-proton antiporter mexB, that govern the expression of MexAB-OprM pump (Fig. 3). A mexR gene mutation V126E detected in exoU carrying strains, was reported to cause overexpression of the MexAB-oprM pump and thus increase the resistance to fluoroquinolones [42,46]. Also, nalC mutations, G71E and S209R were detected mostly in all the strains. Interestingly, three novel strain-specific mutations L85P, N130S and G206S of nalD, a secondary repressor of the MexAB-OprM system, were detected in BK6, BK4 and BK2, respectively. A study showed that mutations in the nalD gene might overexpress mexAB-OprM efflux system [47]; however, further studies are required to confirm their role in clinical outcome.

3.4.3. Virulence factors

We detected several virulence factors and systems including T3SS, Type IV Pili and Fimbrial biogenesis, Pyoverdin biosynthesis, Phenazine biosynthesis, Alginate regulation, Flagellar biosynthesis, Paerucumarin biosynthesis, lasA protease precursor, Twitching motility and Type VI Secretion System in all five genomes of this study and compared them with other keratitis strains (Fig. 4). Among the T3SS toxins, gene encoding exoU was identified in BK3, BK5 and BK6, while exoS, exoT and exoY genes were present in all the strains used in this study. However, the exoS was predicted to be non-functional in BK3 and BK6 due to a very low identity with functional exoS (Fig. 4). Interestingly, a single nucleotide insertion in exoU gene was detected in BK5, which leads to a non-sense mutation (p.Q511X) and possibly truncating 163 amino acids at the C-terminal domain. Furthermore, as shown in Fig. 4, the keratitis strains showed that they carry either exoU or exoS for their virulence. Despite the above fact, we identified that all the exoS carrying strains were non-MDR phenotype, whereas all exoU carrying strains were MDR/XDR except PA39016 due to non-availability of clear antibiotics susceptibility results. Studies reported that strains carrying exoU with resistance to multiple drugs were found to be more virulent and leads to poor clinical outcome [12,48], this was true for BK3 and VRFPA04 but not BK6, suggesting that host-immune response might also play a role in clearing the bacterial infection. Thus, enhancing the host immune

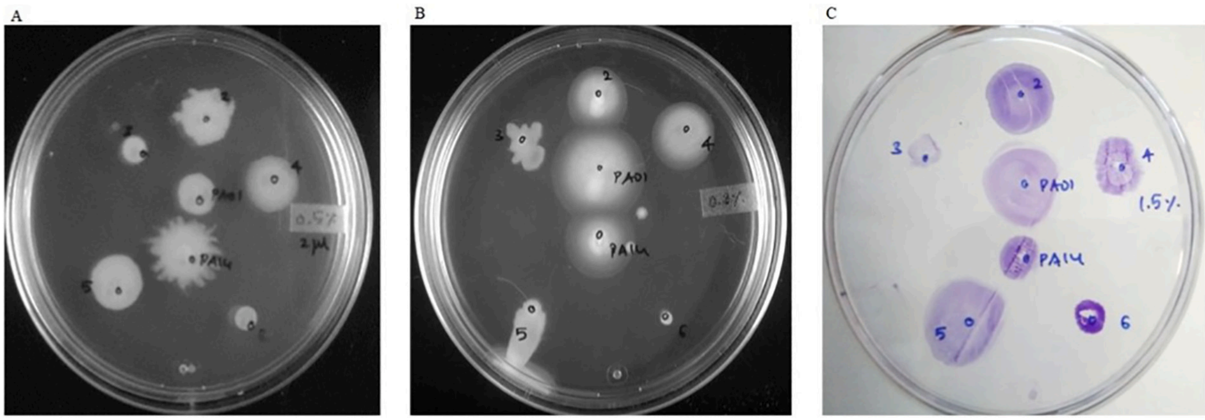

Fig. 5. Motility assays A. swarming B. swimming and C. twitching for *P.aeruginosa* ocular strains of this study and reference strains PAO1 and PA14. The numbers in the culture plates 2–6 represents the ocular strains BK2-BK6 respectively. PA14 represents UCBPP-PA14.

response would help in combating keratitis due to highly-virulent *P. aeruginosa* strains with multiple drug-resistance [49–51].

Among the strains carrying exoS from the poor clinical outcome group, the flagellar genes (fliC, fliD, fliS, flgL, fleP, flgK and fleI/flag) were detected with more than 90% identity with PAO1 in BK1, BK4 and BK5 (Fig. 4). The other exoS strains carrying flagellar genes were PA157 and PA149, isolated from keratitis patients with no reported clinical outcome. Flagellar genes are essential for swimming, twitching and swarming motility of bacteria [52,53]; we performed motility assays in the ocular strains of this study, PAO1 and PA14. Both BK4 and BK5 strains showed all three motilities (Fig. 5). Besides, BK2 also showed all three motilities, suggesting that it might have other motility genes. Indeed, the pilB and pilC genes detected in BK2 with more than 90% sequence identity compared to other strains, were essential for type IV pilus (T4P) expression and twitching motility [54,55]. Nevertheless, among these flagellar genes, fliC and fliD have been reported to induce inflammasome and impairs bacterial clearance [56,57].

3.4.4. Prophages

Several intact prophages were detected in keratitis strains except BK1, PA40 and PA157 (Supplementary Fig. S4). Among them, Psedo_Pf1 and Pseudo_YMC11/02/R656 were detected in at least more than three strains. Pseudo_F10 was detected in BK3 and VRFPA04 strains of poor outcome group, and PA39016. Furthermore, strain-specific phages Pseudo_JD024, Pseudo_MD8 and Pseudo_Phi3 were detected in poor outcome group strains BK3, BK4 and BK5 respectively, suggesting that they may contribute to keratitis pathogenesis.

4. Conclusions

In summary, we noted a larger genome size and more genes in the MDR/XDR group than the non-MDR group. However, 90% of the virulence genes were detected in all strains. Among them, exoU and exoS of the T3SS, the main contributor for keratitis pathogenesis, were detected in MDR/XDR and non-MDR strains, respectively. Previous studies reported that strains carrying exoU with resistance to multiple drugs might lead to poor clinical outcomes [12,48]. However, in this study, one XDR strain carrying exoU was isolated from a patient with good clinical outcomes, suggesting that the host immune response may play a role in *P. aeruginosa* keratitis pathogenesis. Despite the above, the strain-specific genes responsible for virulence and drug-resistance were detected in the poor outcome group. Of the strain-specific genes, fliC and fliD detected in exoS carrying strains reported to altering the host immune response, might impact the clinical outcome. However, further studies using many genomes of pseudomonas keratitis strains with different clinical outcomes are needed to better understand the role of the virulence factors that impact on clinical outcome.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ygeno.2020.08.032.

Ethics approval and consent to participate

This study adhered to the tenets of the Declaration of Helsinki and ethics committee approval was obtained from the Institutional Review Board of the Aravind Eye Care System. Informed consent was obtained from each patient before surgery.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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